Phil Watson

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Education

Worcester Polytechnic Institute, Worcester, MA May 2028

Bachelor of Science of Electrical and Computer Engineering, GPA: 3.95/4.0

Related Coursework: Real Time DSP, Advanced Design with FPGAs, Software Defined Radio, CMOS Fundamentals, Microelectronics II, RF Circuit Design, Software Design for Robotics, Electromechanical Energy Systems

Skills

Electrical: PCB design, firmware, sensors, robotics, FPGAs, circuit design and troubleshooting

Software: KiCad, C++, Python, GitHub, Linux, Helix, Jira, Vivado, SolidWorks, Fusion 360, and STM32CubeMX

Experience

Control Path Intern, Nasuni, Boston, MA June 2026 – August 2026

- Worked in a small team of experienced software engineers designing and programming the orchestration layer of a greenfield cloud storage product in Python and Kubernetes focusing on scalability and high availability
- Worked in a fast paced Agile environment, contributing the design and implementation of the control plane
- Learned from engineers with multiple decades in software engineering about general engineering practices

Service Writer and Spot Mechanic, Landry's Bicycles, Boston, MA May 2022 – May 2026

- Worked with customers by diagnosing mechanical issues, communicating with them about their needs, creating work orders, and using technical problem-solving to overcome obstacles
- Repaired bicycles for customers at work and during community events, such as the Pan-Mass Challenge

Design and Build Captain, AHS Robotics Club, Arlington, MA September 2022 – June 2024

- Managed design and building of the AHS Robotics Club's robot for the FIRST Tech Challenge
- Translated ideas into CAD with Fusion 360 by designing the technical details required for the ideas to work
- Mentored and taught members technical skills, such as designing, modeling, wiring, and building robots while developing leadership, teamwork, and problem-solving skills

Projects

Charger controller for FSAE EV Race Car, Worcester, MA August 2026 – Present

- Designing and implementing a charger control board to interface with a Tesla On-Board Charger, and car's electrical systems, including it's CAN bus, hardware shutdown loop, and high voltage connection to it's battery
- Created design requirements from FSAE rules and team goals, specced components, created schematic and layed out PCB with KiCad, and will program firmware in C++

Embedded Sensor Integration for FSAE EV Race Car, Worcester, MA March 2025 – June 2025

- Designed and implemented an air pressure sensor array to evaluate aerodynamic surfaces on the car in KiCad. Sensors were connected in a UART daisy chain and integrated into the car's CAN bus.
- Designed and implemented wheel speed sensors and a steering angle sensor using C and SolidWorks

Snake on FPGA, Worcester, MA September 2025 – October 2025

- Implemented a VGA controller and game logic in Verilog on a Basys-3 Artix-7 FPGA to display the snake game on a VGA monitor

Cylinder Clock, Cambridge, MA December 2017 – August 2024

- Designed and built a clock that displays time using three vertically stacked machined aluminum cylinders that were prototyped, designed, and CNC milled using Fusion 360

Activities

- **Electrical Team Member**, Goat Fast Racing(FSAE) March 2025 – Present
- **Vice President and Lab Monitor,** WPI CollabLab March 2024 – Present
- **President**, Cheese Club March 2026 – Present
- **Darkroom Manager**, WPI Photo Club February 2025 – Present